

Explanation

❑ Why didn't we create a Feature Importance visualization for the Neural Network model?

Feature Importance was only calculated for the **Random Forest** because tree-based models naturally provide feature importance scores.

Random Forest determines how important each feature is based on how much it contributes to reducing impurity in the decision trees. This allows us to visualize feature importance easily.

However, the **Multilayer Perceptron** (Neural Network) does not provide a built-in feature importance method. Neural networks learn through multiple hidden layers and complex weight combinations, which makes direct feature importance harder to interpret.

Therefore, feature importance visualization is typically performed using tree-based models like Random Forest.

❑ Did we add the Delay feature to the Neural Network model?

Yes. The **Delay** feature was created once and added to the dataset before splitting the data into training and testing sets.

Since the feature was added to the feature matrix **X**, both models used it:

- Random Forest
- Neural Network

The only extra step applied to the Neural Network was **feature scaling using StandardScaler**, but the same features (including Delay) were used for training.

Therefore, the Delay feature was included in **both models**.

❓ Which model has the higher accuracy?

The **Random Forest model** achieved the higher accuracy.

Model	Accuracy
Random Forest	Higher
Neural Network	Slightly Lower

Random Forest predicted all test samples correctly, which resulted in the highest accuracy on this dataset.

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Random Forest Model
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Best Random Forest params: {'max_depth': None, 'min_samples_split': 2, 'n_estimators': 100}
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Accuracy: 1.0
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Warning: Warning
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Best Neural Network params: {'activation': 'relu', 'alpha': 0.0001, 'hidden_layer_sizes': (50,,)}
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```
Accuracy: 0.9666666666666667
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❓ In the Confusion Matrix, is it normal for them to be different? And shouldn't the Actual values be the same?

Yes, it is **normal for the confusion matrices to be slightly different**, because each model makes predictions differently.

However, the **Actual values must remain the same**, because both models are evaluated using the **same test dataset** created during the train-test split.

Random Forest Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	35	0
Actual 1	0	25

Neural Network Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	34	1
Actual 1	1	24

Checking the Actual values

Actual Class 0:

- Random Forest $\rightarrow 35 + 0 = 35$
- Neural Network $\rightarrow 34 + 1 = 35$

Actual Class 1:

- Random Forest $\rightarrow 0 + 25 = 25$
- Neural Network $\rightarrow 1 + 24 = 25$

So the **Actual values are exactly the same in both models.**

The difference only appears in the **predicted values**, because each model learns patterns differently.

For example, the Neural Network misclassified **two samples**, while Random Forest classified all samples correctly.

Conclusion

- Feature importance was visualized only for Random Forest because it provides built-in importance scores.
- The Delay feature was used by both models.
- Random Forest achieved the highest accuracy.
- Confusion matrices can differ between models, but the total number of actual samples must remain the same.